

Statement of participation

Syed kamil Taqi Zaidi

has completed the free course including any mandatory tests for:

Unsolved problems in cosmology

This free 6-hour course introduced the major unsolved problems in cosmology, and how they may be solved in the future.

Issue date: 8 October 2026



www.open.edu/openlearn

This statement does not imply the award of credit points nor the conferment of a University Qualification.
This statement confirms that this free course and all mandatory tests were passed by the learner.

Please go to the course on OpenLearn for full details:
<https://www.open.edu/openlearn/science-maths-technology/unsolved-problems-cosmology/content-section-0>

COURSE CODE: **S385_1**

Unsolved problems in cosmology

<https://www.open.edu/openlearn/science-maths-technology/unsolved-problems-cosmology/content-section-0>

Course summary

The science of cosmology has been very successful in explaining a wide range of observational evidence about the Universe, from the detailed properties of the cosmic microwave background radiation to the distributions of galaxies, and the elemental abundances in astrophysical environments. However, there are several major unknowns in cosmological theory, including the nature of dark matter and dark energy. These gaps in knowledge are frustrating for anyone who would like to understand the nature of the Universe. In this free course, *Unsolved problems in cosmology*, you will learn about what the current unsolved problems are and about the possible ways in which they may be solved in the future.

Learning outcomes

By completing this course, the learner should be able to:

- understand the basis of modern cosmology based on the hot big bang
- summarise the main candidate dark matter particles, and the prospects for directly detecting them
- compare models to explain the late-time acceleration of the Universe's expansion, i.e. dark energy and the cosmological constant
- discuss observational prospects for understanding the nature of dark energy
- explain the theoretical problems that led to the theory of inflation, how an early inflationary period solves them and how researchers are investigating inflation.

Completed study

The learner has completed the following:

Section 1

Cosmology today

Section 2

Is modern cosmology fundamentally wrong?

Section 3

The nature of dark matter

Section 4

The nature of dark energy

Section 5

Inflation

Section 6

Quiz

Section 7

Conclusion